Lab 3

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4-25-2026

CDA 3203 Computer Logic Design

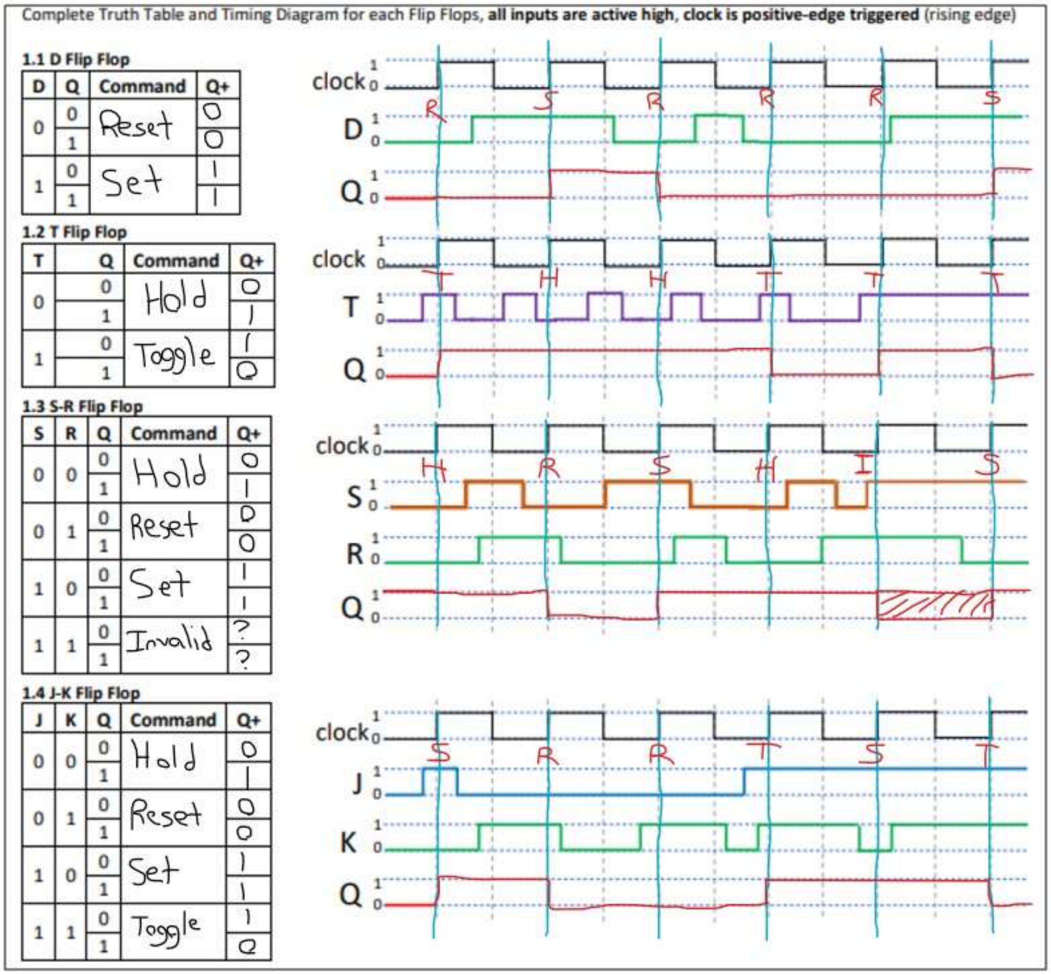
Spring 2026

Dr. Maria Petrie

Florida Atlantic University

3.0) Handwork (40 pts)____________________

Fill in the Diagrams and Truth Tables below, you can delete the image below and paste yours in. Or if you can draw/annotate on top of the image below – using OneNote or another notetaking app.



1.1 D Flip Flop

D	Q	Command	Q+
0	0	Reset	0
	1		0
1	0	Set	1
	1		1

1.2 T Flip Flop

T	Q	Command	Q+
0	0	Hold	0
	1		1
1	0	Toggle	1
	1		0

1.3 S-R Flip Flop

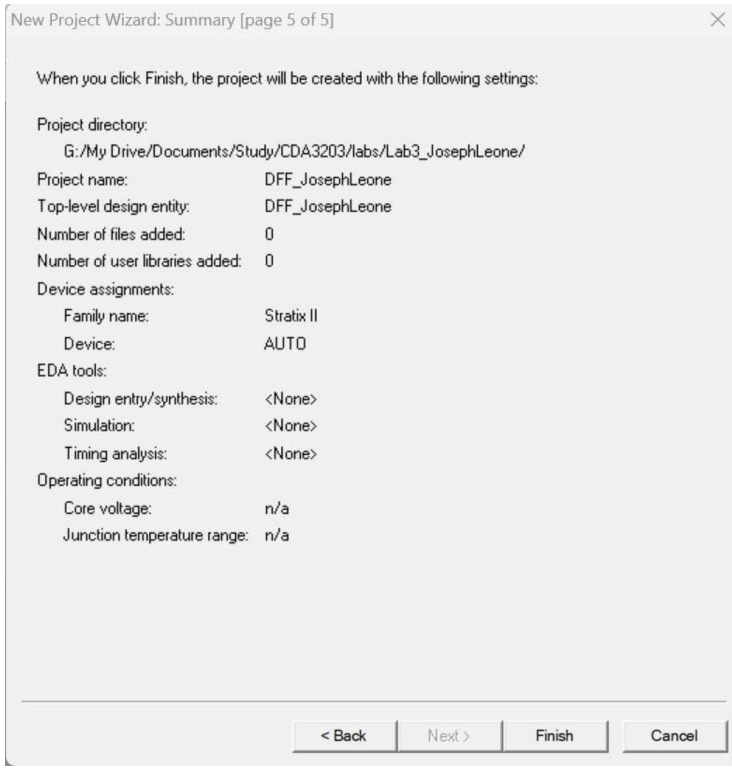
S	R	Q	Command	Q+
0	0	0	Hold	0
		1		1
0	1	0	Reset	0
		1		0
1	0	0	Set	1
		1		1
1	1	0	Invalid	?
		1		?

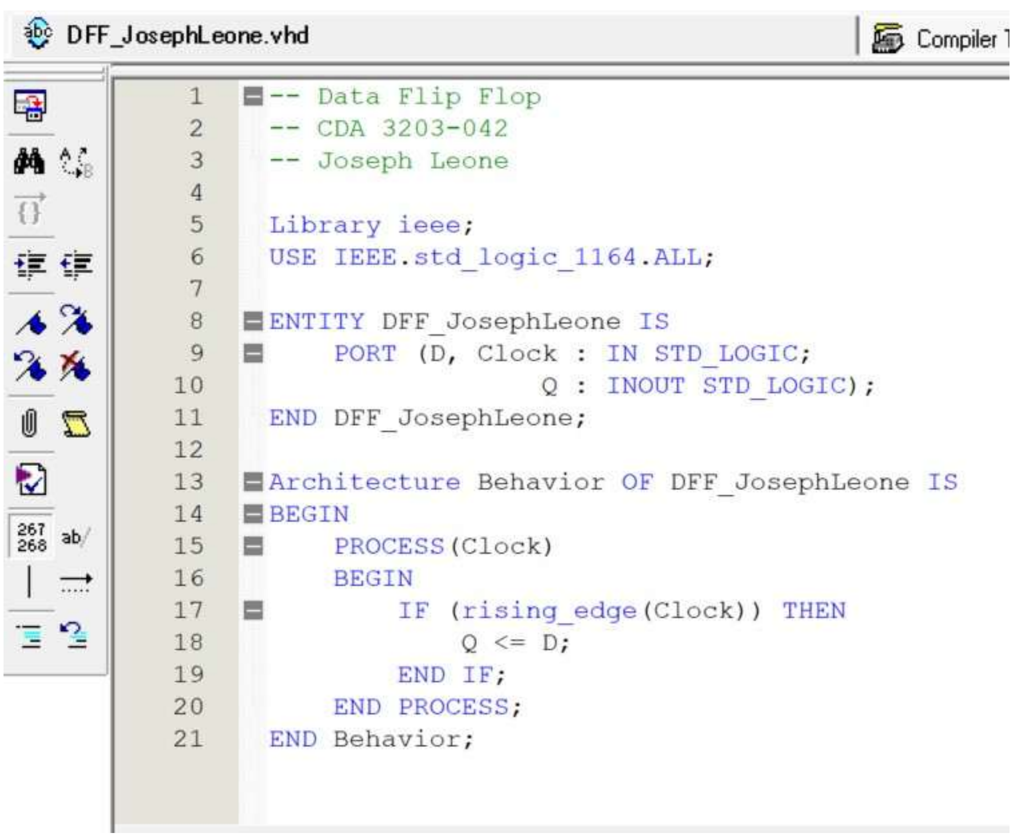
1.4 J-K Flip Flop

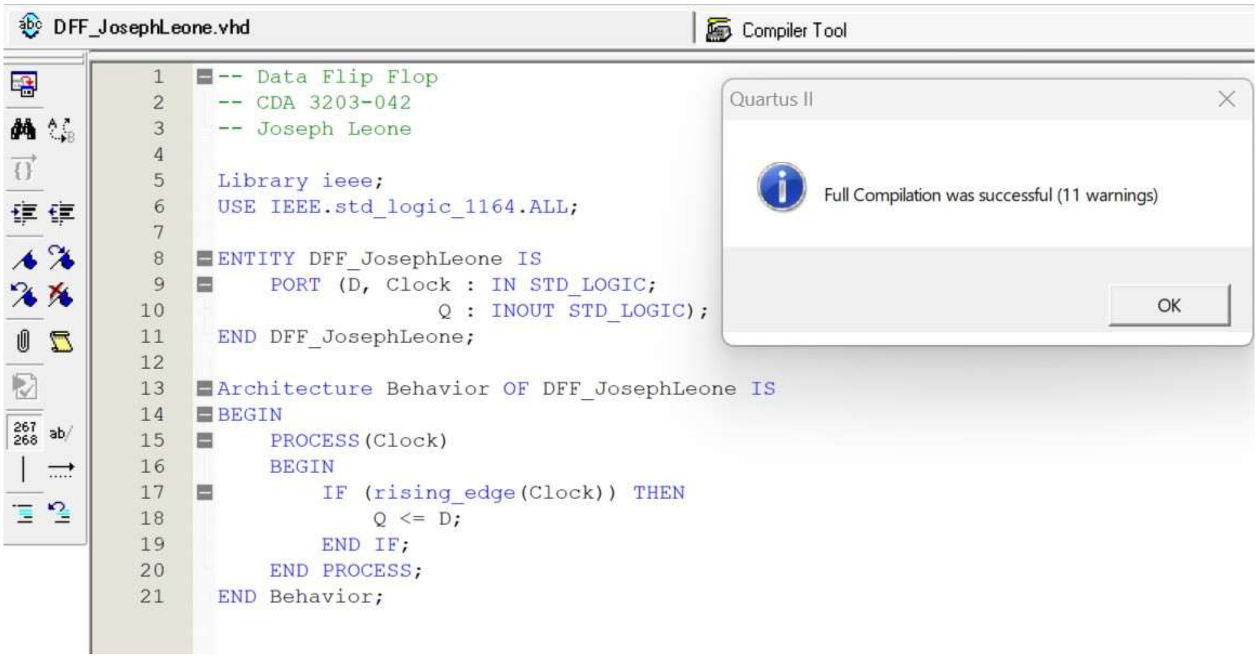
J	K	Q	Command	Q+
0	0	0	Hold	0
		1		1
0	1	0	Reset	0
		1		0
1	0	0	Set	1
		1		1
1	1	0	Toggle	1
		1		0

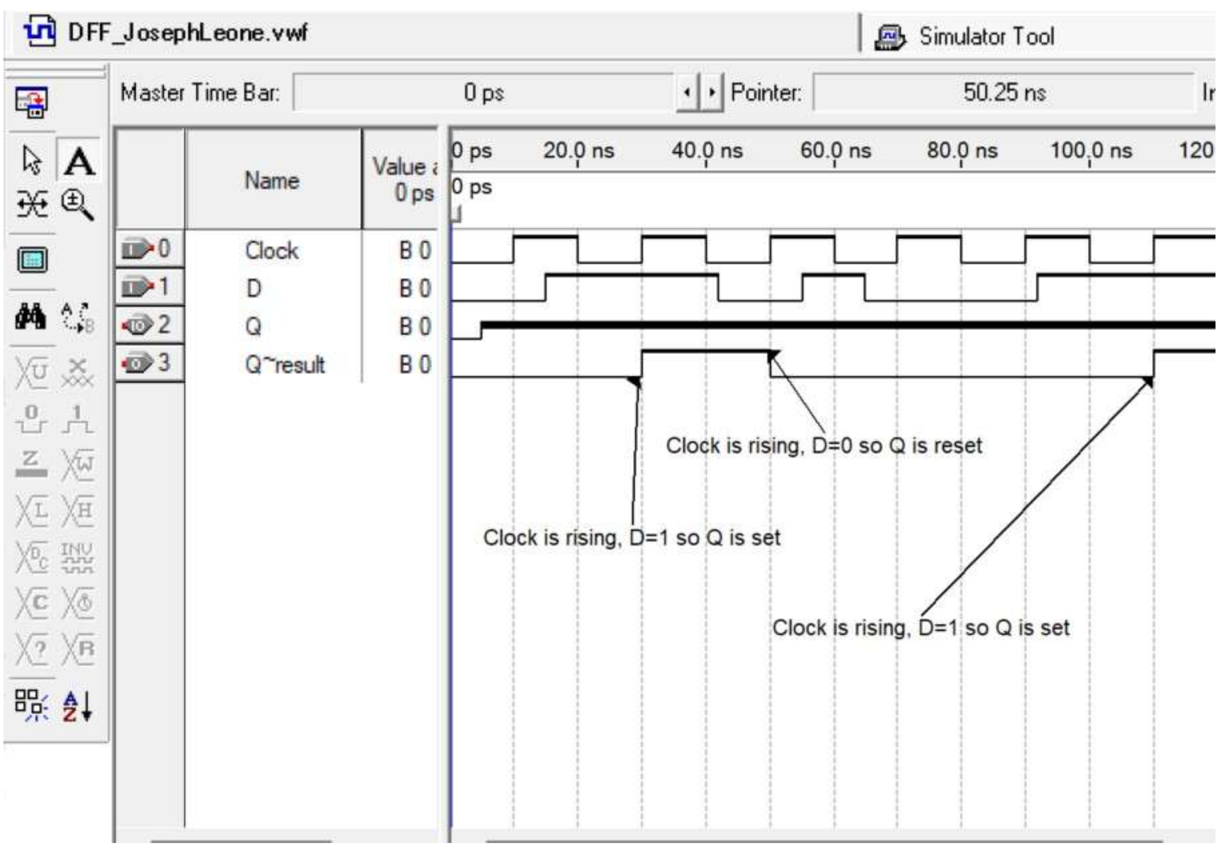
3.1) Simulating a Data Flip Flop (DFF)(20pts) ____________________

Dr. Petrie gives you the code for this Entity. Watch the videos!

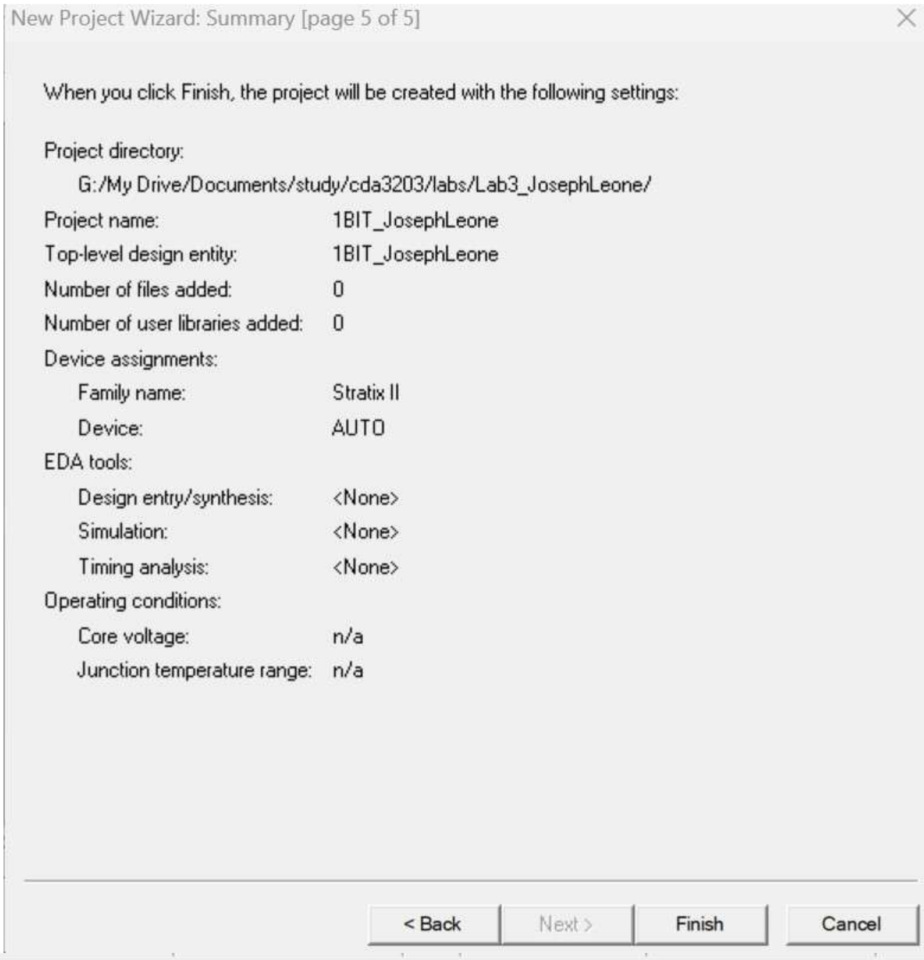


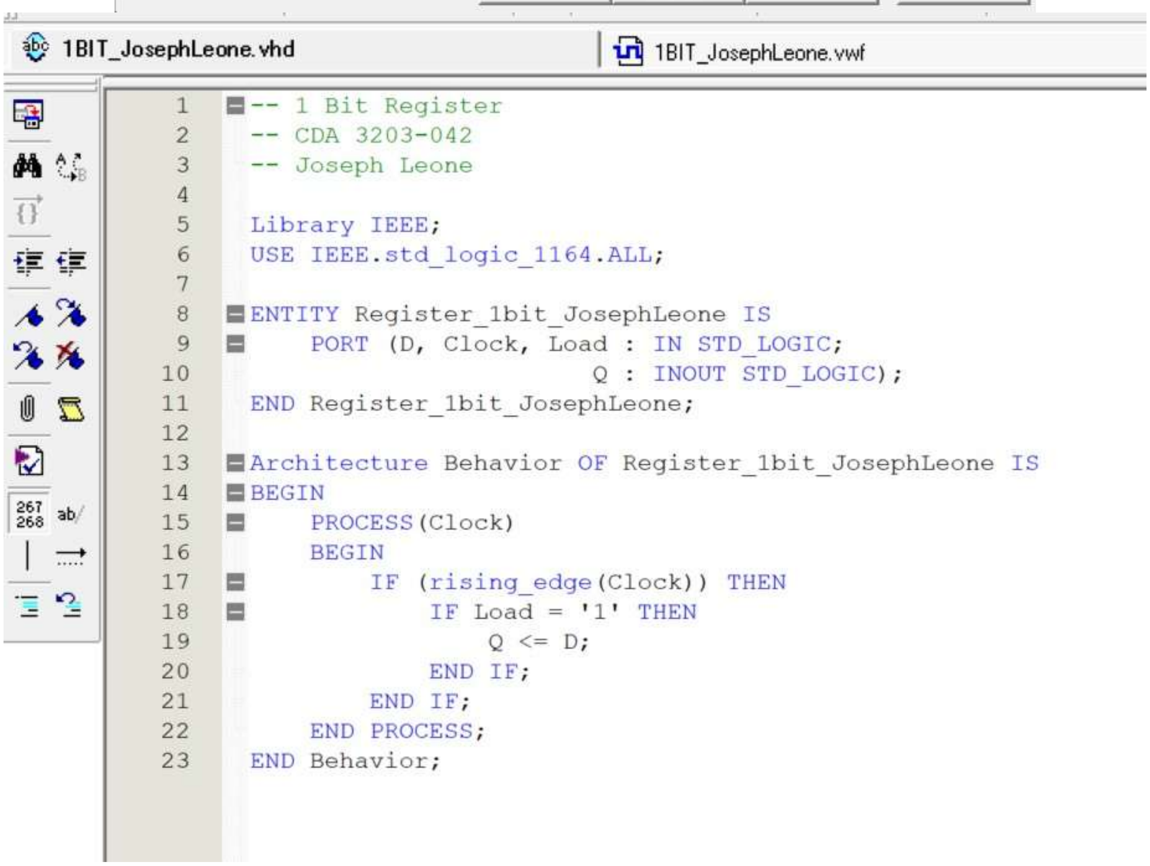



DFF_JosephLeone.vhd
Compiler Tool
-- Data Flip Flop
-- CDA 3203-042
-- Joseph Leone
Library ieee;
USE IEEE.std_logic_1164.ALL;
ENTITY DFF_JosephLeone IS
PORT (D, Clock : IN STD_LOGIC;
Q : INOUT STD_LOGIC);
END DFF_JosephLeone;
Architecture Behavior OF DFF_JosephLeone IS
BEGIN
PROCESS(Clock)
BEGIN
IF (rising_edge(Clock)) THEN
Q <= D;
END IF;
END PROCESS;
END Behavior;
Quartus II
Full Compilation was successful (11 warnings)
OK


DFF_JosephLeone.vwf
Simulator Tool
Master Time Bar:
0 ps
Pointer:
50.25 ns
Name
Value at 0 ps
0 ps
20.0 ns
40.0 ns
60.0 ns
80.0 ns
100.0 ns
120
0
Clock
B 0
1
D
B 0
2
Q
B 0
3
Q~result
B 0
Clock is rising, D=0 so Q is reset
Clock is rising, D=1 so Q is set
Clock is rising, D=1 so Q is set

3.2) Designing and simulating a 1-Bit Register (40pts) ___________



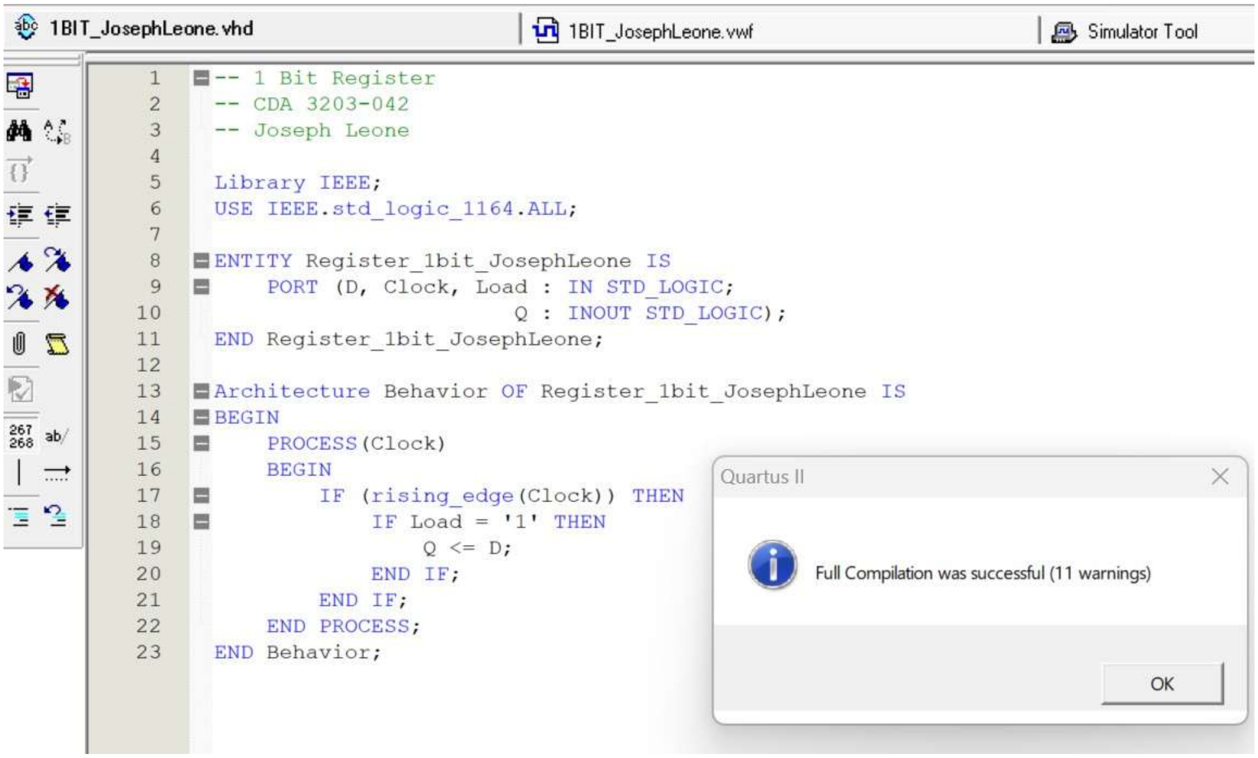


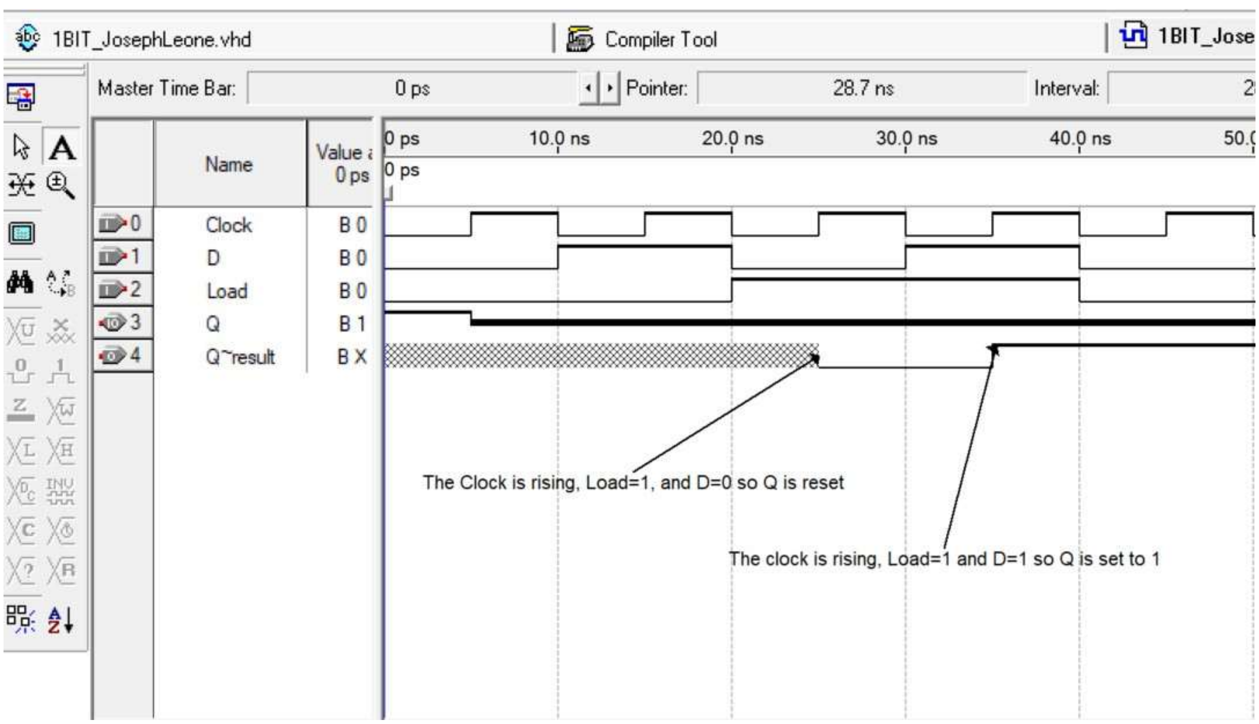

```vhdl
-- 1 Bit Register
-- CDA 3203-042
-- Joseph Leone

Library IEEE;
USE IEEE.std_logic_1164.ALL;

ENTITY Register_1bit_JosephLeone IS
    PORT (D, Clock, Load : IN STD_LOGIC;
                       Q : INOUT STD_LOGIC);
END Register_1bit_JosephLeone;

Architecture Behavior OF Register_1bit_JosephLeone IS
BEGIN
    PROCESS(Clock)
    BEGIN
        IF (rising_edge(Clock)) THEN
            IF Load = '1' THEN
                Q <= D;
            END IF;
        END IF;
    END PROCESS;
END Behavior;
```

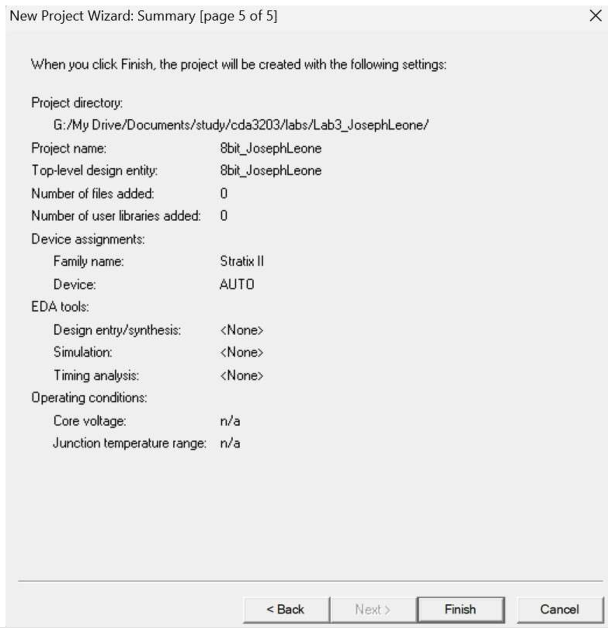

1BIT_JosephLeone.vhd
1BIT_JosephLeone.vwf
Simulator Tool
-- 1 Bit Register
-- CDA 3203-042
-- Joseph Leone
Library IEEE;
USE IEEE.std_logic_1164.ALL;
ENTITY Register_1bit_JosephLeone IS
PORT (D, Clock, Load : IN STD_LOGIC;
Q : INOUT STD_LOGIC);
END Register_1bit_JosephLeone;
Architecture Behavior OF Register_1bit_JosephLeone IS
BEGIN
PROCESS(Clock)
BEGIN
IF (rising_edge(Clock)) THEN
IF Load = '1' THEN
Q <= D;
END IF;
END IF;
END PROCESS;
END Behavior;
Quartus II
Full Compilation was successful (11 warnings)
OK

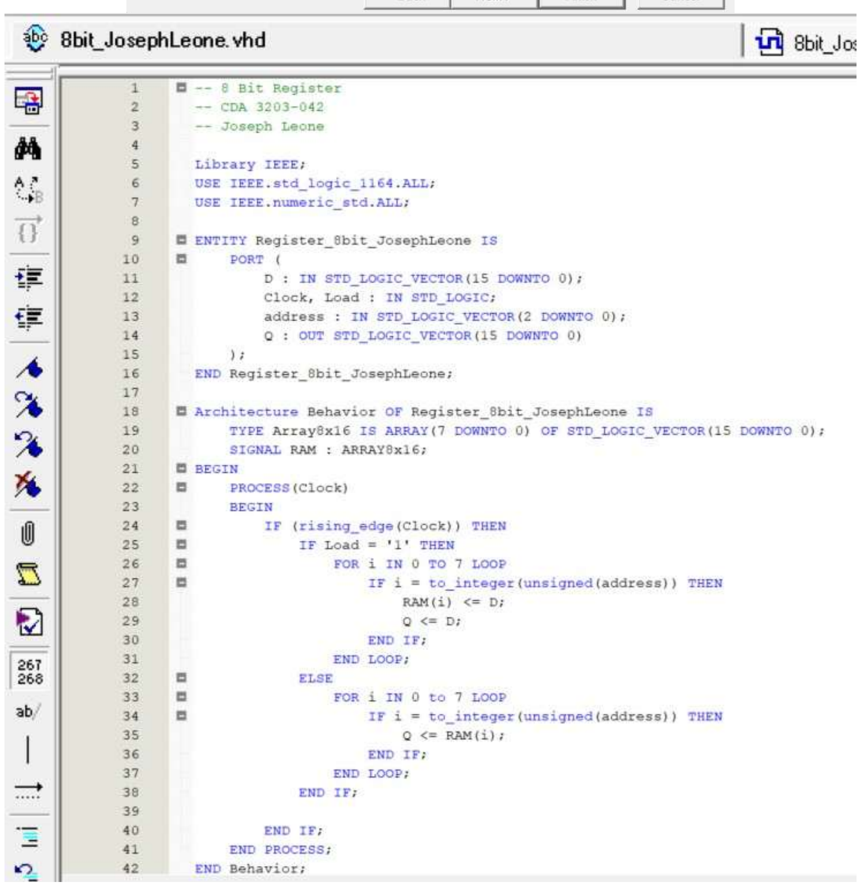

1BIT_JosephLeone.vhd
Compiler Tool
Master Time Bar:
0 ps
Pointer:
28.7 ns
Interval:
Name
0 ps
10.0 ns
20.0 ns
30.0 ns
40.0 ns
Clock
B 0
D
B 0
Load
B 0
Q
B 1
Q~result
B X
The Clock is rising, Load=1, and D=0 so Q is reset
The clock is rising, Load=1 and D=1 so Q is set to 1

3.3) EXTRA CREDIT ENTITY

Designing and Simulating a 8-Bit Register (20pts)______________________________

No Structural code, ONLY behavioral. This is an ACTIVE HIGH entity. You are expected to ANNOTATE the. vwf behavior and why your timing diagram is correct for full credit.






```vhdl
-- 8 Bit Register
-- CDA 3203-042
-- Joseph Leone

Library IEEE;
USE IEEE.std_logic_1164.ALL;
USE IEEE.numeric_std.ALL;

ENTITY Register_8bit_JosephLeone IS
    PORT (
        D : IN STD_LOGIC_VECTOR(15 DOWNTO 0);
        Clock, Load : IN STD_LOGIC;
        address : IN STD_LOGIC_VECTOR(2 DOWNTO 0);
        Q : OUT STD_LOGIC_VECTOR(15 DOWNTO 0)
    );
END Register_8bit_JosephLeone;

Architecture Behavior OF Register_8bit_JosephLeone IS
    TYPE Array8x16 IS ARRAY(7 DOWNTO 0) OF STD_LOGIC_VECTOR(15 DOWNTO 0);
    SIGNAL RAM : ARRAY8x16;
BEGIN
    PROCESS(Clock)
    BEGIN
        IF (rising_edge(Clock)) THEN
            IF Load = '1' THEN
                FOR i IN 0 TO 7 LOOP
                    IF i = to_integer(unsigned(address)) THEN
                        RAM(i) <= D;
                        Q <= D;
                    END IF;
                END LOOP;
            ELSE
                FOR i IN 0 to 7 LOOP
                    IF i = to_integer(unsigned(address)) THEN
                        Q <= RAM(i);
                    END IF;
                END LOOP;
            END IF;
        END IF;
    END PROCESS;
END Behavior;
```